```
→  assignment-5 ./schedule input.1 FCFS
t=0-12: run pid 0
t=12: pid 0 finished
t=12-16: run pid 1
t=16: pid 1 finished
t=16-17: run pid 2
t=17: pid 2 finished
t=17-19: run pid 3
t=19: pid 3 finished

PID Arrival Start End Running Waiting
  0       0     0  12      12       0
  1       2    12  16       4      10
  2       3    16  17       1      13
  3       4    17  19       2      13
Average Waiting Time: 9
→  assignment-5 ./schedule input.2 FCFS
t=0-2: run pid 1
t=2: pid 1 finished
t=2: idle
t=3: idle
t=4: idle
t=5-8: run pid 0
t=8: pid 0 finished
t=8: idle
t=9: idle
t=10: idle
t=11: idle
t=12: idle
t=13: idle
t=14: idle
t=15: idle
t=16: idle
t=17: idle
t=18: idle
t=19: idle
t=20-21: run pid 2
t=21: pid 2 finished

PID Arrival Start End Running Waiting
  0       5     5   8       3       0
  1       0     0   2       2       0
  2      20    20  21       1       0
Average Waiting Time: 0
→  assignment-5 ./schedule input.3 FCFS
t=0-7: run pid 0
t=7: pid 0 finished
t=7-10: run pid 1
t=10: pid 1 finished
t=10-15: run pid 2
t=15: pid 2 finished
t=15-16: run pid 3
t=16: pid 3 finished
t=16-20: run pid 4
t=20: pid 4 finished

PID Arrival Start End Running Waiting
  0       0     0   7       7       0
  1       1     7  10       3       6
  2       2    10  15       5       8
  3       3    15  16       1      12
  4       4    16  20       4      12
Average Waiting Time: 7.6
```

FCFS

```
→  assignment-5 ./schedule input.1 RR 2
t=0-2: run pid 0
t=2-4: run pid 1
t=4-6: run pid 0
t=6-7: run pid 2
t=7: pid 2 finished
t=7-9: run pid 3
t=9: pid 3 finished
t=9-11: run pid 1
t=11: pid 1 finished
t=11-13: run pid 0
t=13-15: run pid 0
t=15-17: run pid 0
t=17-19: run pid 0
t=19: pid 0 finished

PID Arrival Start End Running Waiting
  0       0     0  19      12       7
  1       2     2  11       4       5
  2       3     6   7       1       3
  3       4     7   9       2       3
Average Waiting Time: 4.5
→  assignment-5 ./schedule input.2 RR 2
t=0-2: run pid 1
t=2: pid 1 finished
t=2: idle
t=3: idle
t=4: idle
t=5-7: run pid 0
t=7-8: run pid 0
t=8: pid 0 finished
t=8: idle
t=9: idle
t=10: idle
t=11: idle
t=12: idle
t=13: idle
t=14: idle
t=15: idle
t=16: idle
t=17: idle
t=18: idle
t=19: idle
t=20-21: run pid 2
t=21: pid 2 finished

PID Arrival Start End Running Waiting
  0       5     5   8       3       0
  1       0     0   2       2       0
  2      20    20  21       1       0
Average Waiting Time: 0
→  assignment-5 ./schedule input.3 RR 2
t=0-2: run pid 0
t=2-4: run pid 1
t=4-6: run pid 2
t=6-8: run pid 0
t=8-9: run pid 3
t=9: pid 3 finished
t=9-11: run pid 4
t=11-12: run pid 1
t=12: pid 1 finished
t=12-14: run pid 2
t=14-16: run pid 0
t=16-18: run pid 4
t=18: pid 4 finished
t=18-19: run pid 2
t=19: pid 2 finished
t=19-20: run pid 0
t=20: pid 0 finished

PID Arrival Start End Running Waiting
  0       0     0  20       7      13
  1       1     2  12       3       8
  2       2     4  19       5      12
  3       3     8   9       1       5
  4       4     9  18       4      10
Average Waiting Time: 9.6
```

RR (time quantum = 2)

```
→  assignment-5 ./schedule input.1 SJF
t=0-12: run pid 0
t=12: pid 0 finished
t=12-13: run pid 2
t=13: pid 2 finished
t=13-15: run pid 3
t=15: pid 3 finished
t=15-19: run pid 1
t=19: pid 1 finished

PID Arrival Start End Running Waiting
  0       0     0  12      12       0
  1       2    15  19       4      13
  2       3    12  13       1       9
  3       4    13  15       2       9
Average Waiting Time: 7.75
→  assignment-5 ./schedule input.2 SJF
t=0-2: run pid 1
t=2: pid 1 finished
t=2: idle
t=3: idle
t=4: idle
t=5-8: run pid 0
t=8: pid 0 finished
t=8: idle
t=9: idle
t=10: idle
t=11: idle
t=12: idle
t=13: idle
t=14: idle
t=15: idle
t=16: idle
t=17: idle
t=18: idle
t=19: idle
t=20-21: run pid 2
t=21: pid 2 finished

PID Arrival Start End Running Waiting
  0       5     5   8       3       0
  1       0     0   2       2       0
  2      20    20  21       1       0
Average Waiting Time: 0
→  assignment-5 ./schedule input.3 SJF
t=0-7: run pid 0
t=7: pid 0 finished
t=7-8: run pid 3
t=8: pid 3 finished
t=8-11: run pid 1
t=11: pid 1 finished
t=11-15: run pid 4
t=15: pid 4 finished
t=15-20: run pid 2
t=20: pid 2 finished

PID Arrival Start End Running Waiting
  0       0     0   7       7       0
  1       1     8  11       3       7
  2       2    15  20       5      13
  3       3     7   8       1       4
  4       4    11  15       4       7
Average Waiting Time: 6.2
```

SJF

How to build

```
clang++ -o schedule schedule.cpp
```

How to run

```
./schedule.cpp [input file] [Schedule method] [time quantum]
```